# Think like a computer

**A programmer must learn to think like a computer. All tasks must be broken down into small chunks so that they are easy to follow and impossible to get wrong.**

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## Thinking like a robot

Imagine a café where the waiter is a robot. The robot has a simple computer brain, and needs to be told how to get from the café kitchen to serve food to diners seated at tables. First the process has to be broken down into simple tasks the computer can understand.

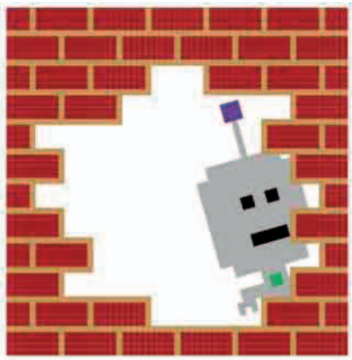
LINGO

### Algorithm

An algorithm is a set of simple instructions for performing a task. A program is an algorithm that has been translated into a language that computers can understand.

**1 Waiter robot program 1**
Using this program the robot grabs the food from the plate, crashes straight through the kitchen wall into the dining area, and puts the food on the floor. This algorithm wasn't detailed enough.

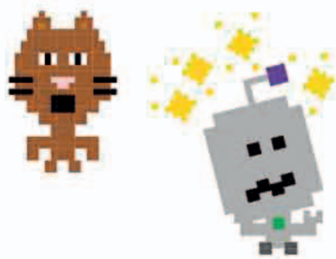
1. Pick up food
2. Move from kitchen to diner's table
3. Put food down



◁ **Disaster!**
The instructions weren't clear: we forgot to tell the robot to use the door. It might seem obvious to humans but computers can't think for themselves.

**2 Waiter robot program 2**
This time we've told the robot waiter to use the kitchen door. It makes it through the door, but then hits the café cat, trips, and smashes the plate on the floor.

1. Pick up a plate with food on it
2. Move from kitchen to diner's table by:
    - Move to door between kitchen and dining area
    - Move from door to the table
3. Put plate down on the table in front of the diner



△ **Still not perfect**
The robot doesn't know how to deal with obstacles like the cat. The program needs to give the robot even more detailed instructions so it can move around safely.